Smart IoT-Based Hazardous Gas Detection and Monitoring System for Ghee & Rica Food Manufacturer

Rationale

Ensuring workplace safety in food manufacturing facilities is essential to protect employees, maintain product quality, and prevent operational disruptions. Ghee & Rica Food Manufacturer operates in an environment where gas-powered equipment and processing systems may pose potential risks if gas leaks occur. Undetected hazardous gas concentrations can result in serious health issues, fire hazards, and production downtime.

Traditional monitoring systems often rely on manual inspections or independent detectors that lack centralized monitoring and real-time alert capabilities. These limitations may delay response time and increase safety risks.

To address these concerns, the proposed Smart IoT-Based Hazardous Gas Detection and Monitoring System is designed to continuously monitor gas levels in real time. The system will automatically detect abnormal gas levels, trigger alerts, and notify responsible personnel. Through a centralized dashboard, administrators can monitor environmental conditions, review historical records, and configure safety thresholds to ensure a safer and more efficient working environment at Ghee & Rica Food Manufacturer.

Project Purpose

The proposed project aims to design and implement a smart gas detection and monitoring system that enhances workplace safety, improves hazard response time, and provides centralized monitoring and reporting within Ghee & Rica Food Manufacturer.

Research Objectives

- To develop a hardware prototype using gas sensors capable of detecting hazardous gas levels.
- To design a real-time monitoring dashboard that displays current gas readings.
- To implement an automated alert system that notifies Factory Staff and Safety Officers when unsafe gas levels are detected.
- To create a centralized dashboard for monitoring gas levels, viewing historical data, and generating safety reports.

Scope and Limitation of the Study

Scope

The system will focus on monitoring hazardous gas levels within selected operational areas of Ghee & Rica Food Manufacturer. The platform will include real-time gas detection, automatic alert notifications, dashboard monitoring, report generation, and configuration of gas threshold levels.

Limitations

- The system can only monitor areas where gas sensors are physically installed and does not cover the entire facility unless additional sensors are deployed.
- The system requires stable power supply and internet or local network connectivity for real-time monitoring and alert transmission.
- The system functions as an early warning and monitoring tool only and does not replace official safety procedures or emergency response systems.

Benefits/Anticipated Outcomes

Factory Staff: Provides immediate notification in case of hazardous gas detection, enabling faster response and improved workplace safety.

Safety Officer: Enables centralized monitoring, access to historical data, report generation, and configuration of safety thresholds for improved safety management.

Methodology

This study will use the Agile Software Development Framework to guide the design, development, testing, and continuous improvement of the system.

Requirement Gathering: Interviews and consultations will be conducted with management and safety personnel of Ghee & Rica Food Manufacturer to determine safety requirements and system specifications.

System Development: The mobile app will be designed and the sensor hardware set up with manual control functions.

Testing and Deployment: Testing will be conducted on-site to verify that the sensors are accurate and the app responds properly.

Evaluation and Improvement: We will collect feedback from users to make the app more reliable and easier to use.

Proponents:

Jhon Paul C. Bondoc / 2023312239@pampangastateu.edu.ph/ **DMC-BSINFO-3B**

John Emmanuel P. Hipolito / 2023309197@pampangastateu.edu.ph / **DMC-BSINFO-3B**

Guiller T. Sumague / 2022305444@pampangstateu.edu.ph/ **DMC-BSINFO-3B**

Alyssa G. Esguerra / 2023313832@pampangastateu.edu.ph / **DMC-BSINFO-3B**

Tristan De Guzman / 2022307094@pampangastateu.edu.ph**DMC-BSINFO-3B**